SHRI G.S. INTITUTE OF TECHNOLOGY AND SCIENCE
INDORE (M.P.)

Department of Computer Technology and Applications

MCA First Year Semester-II

Session:- January 2026 – June 2026

Subject:- Computer Network

Lab Assignment (Section B)

Submitted To:-

Prof. Sukanya Sinha

Submitted By:-

Saloni Patidar

(0801CA251122)

Assignment-1

S.No.	INDEX	Page No.	Sign
1	Study on different types of cables and connectors used in computer networking.	1	
2	Write process of crimping of LAN cable.	5	
3	What is cross connection and straight connection in networking device.	6	

0801CA251105

Q.1) Study on different types of cables and connectors used in computer networking.

Ans. 1) An ethernet cable allows the user to connect their devices such as computers, mobile phones, routers, etc., to a Local Area Network (LAN) that will allow a user to have internet access, and able to communicate with each other through a wired connection.

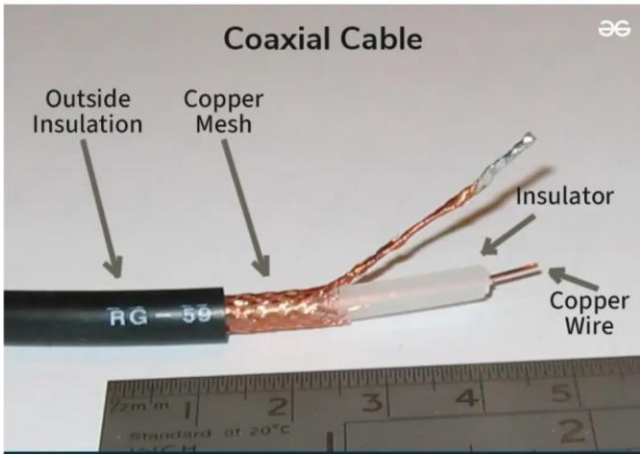
Types of Ethernet Cables

Mainly there are three types of ethernet cables used in LANs

- Coaxial Cables
- Twisted Pair Cables
- Fiber optic Cables

1. Coaxial Cables

A coaxial cable is used to carry high-frequency electrical signals with low losses. It uses 10Base2 and 10Base5 Ethernet variants. It has a copper conductor in the middle that is surrounded by a dielectric insulator generally made of PVC or Teflon. The dielectric insulator is surrounded by a plaited conducting metallic shield which reduces Electromagnetic Interference of the metal and outside interference and finally, the metallic shield is covered by a plastic covering called a sheath usually made of PVC or some other fire-resistant plastic material. Its maximum transmission speed is 10 Mbps. It is usually used in telephone systems, cable TV, etc.



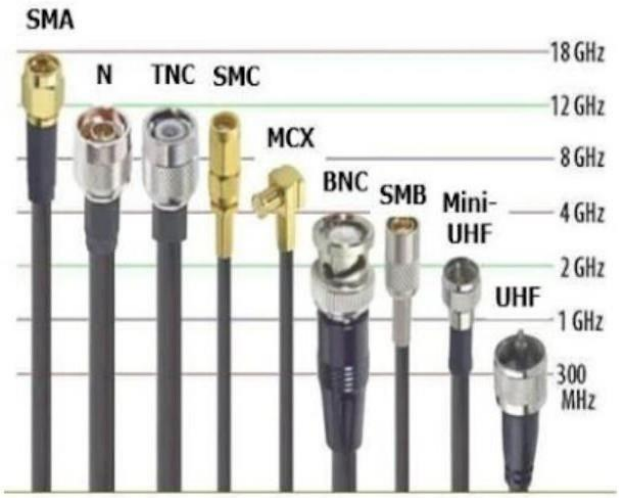
Types of Coaxial Cables

- **Hardline Coaxial Cable: It** is used in applications where high signal strength is required, this type is most commonly used. They are used in internet lines and telephone lines.
- **RG-6 Coaxial Cable:** It is used where better signal quality is required; it has a thicker dielectric insulator, they are used in broadband internet, cable TV, etc.
- **Tri-Axial Cable:** They offer more bandwidth and interference rejection; they use an additional copper braid shield. Commonly used in connecting cameras and cable TVs. Etc.

0801CA251105

Common Types of Connectors Used in Coaxial Cable

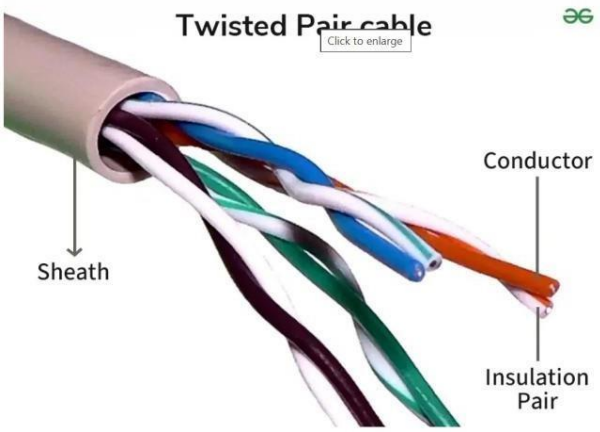
- BNC (Bayonet Neil Concelman): Quick connect/disconnect with a bayonet lug (twist-and-lock), popular for video, test equipment (oscilloscopes), and older networking.
- F Type Connectors: The standard for cable TV, satellite, and cable internet connections in homes, known for its screw-on connection.
- SMA or Subminiature Connector: Used for Wi-Fi, GPS, and mobile devices, offering excellent performance at higher frequencies.
- TNC (Threaded Neil Concelman): A threaded version of BNC, offering better weatherproofing and performance at higher frequencies (up to 12 GHz) for antennas and cellular.



2. Twisted Pair Cable

A twisted pair is a copper wire cable in which two insulated copper wires are twisted around each other to reduce interference or crosstalk. It uses 10BASE-T, 100BASE-T, and some other newer ethernet variants. It uses RJ-45 connectors.

Design: A twisted pair cable usually contains two or more conducting wires either shielded by an insulator or not and, further these twisted pairs of wires are coated for protection from any damage.



0801CA251105

Types of Twisted Pair Cable

- **Shielded Twisted Pair (STP) Cable:** In STP the wires are covered by a copper braid covering or a foil shield, this foil shield adds a layer that protects it against interference leaking into and out of the cable. Hence, they are used for longer distances and higher transmission rates.
- **Unshielded Twisted Pair (UTP) Cable:** Unshielded twisted pair cable is one of the most commonly used cables in computer networks at present time. UTP consists of two insulated copper wires twisted around one another, the twisting of wires helps in controlling interference.

RJ Connectors

RJ stands for Registered Jacks. They are mainly used to connect to different types of networking media. Different types of RJ are available, including RJ11, RJ45, RJ48, RJ61 etc. RJ45 stands for Registered Jack 45 and is the most commonly used connector in wired networks for Twisted Pair Cable. The jacks are mainly used to connect to the Local Area Network (LAN). It was earlier devised for telephones but is now widely used in Ethernet Networking.

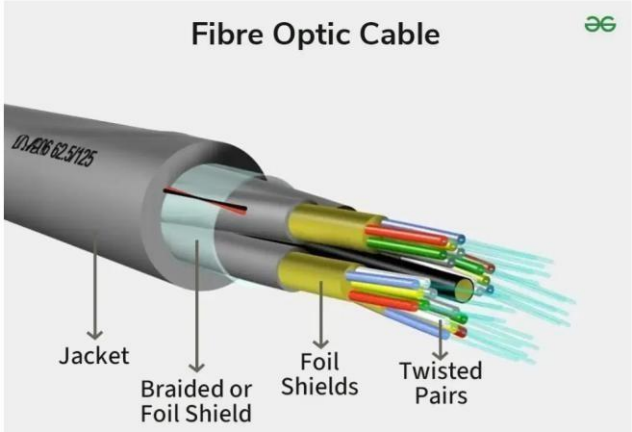
RJ-45 Connector

3. Fiber Optic Cable

Fiber optic cables use optical fibers which are made of glass cores surrounded by several layers of covering material generally made of PVC or Teflon. It transmits data in the form of light signals due to which there are no interference issues in fiber optics. Fiber optics can transmit signals over a very long distance as compared to twisted pairs or coaxial cables. It uses 10BaseF, 100BaseFX, 100BaseBX, 100BaseSX, 1000BaseFx, 1000BaseSX, and 1000BaseBx ethernet variants. Hence, it is capable of carrying information at a great speed.

0801CA251105

Design: An optical fiber consists of a core and a cladding chosen for their total internal reflection due to the difference in refractive index between the two. In real optical fibers, the cladding is usually covered with a layer of acrylate or polyimide polymer. The coating protects the fiber from damage and several layers of protective sheathing, depending on the application are added to form the cable.



Types of Fiber Optics Cable

- **Single-Mode Fiber:** It uses one single ray of light to transmit data. It is used for long-distance transmission.
- **Multi-Mode Fiber:** It uses multiple light rays to transmit data. It is comparatively less expensive.

Common Types of Connectors Used in Fiber Optics Cable

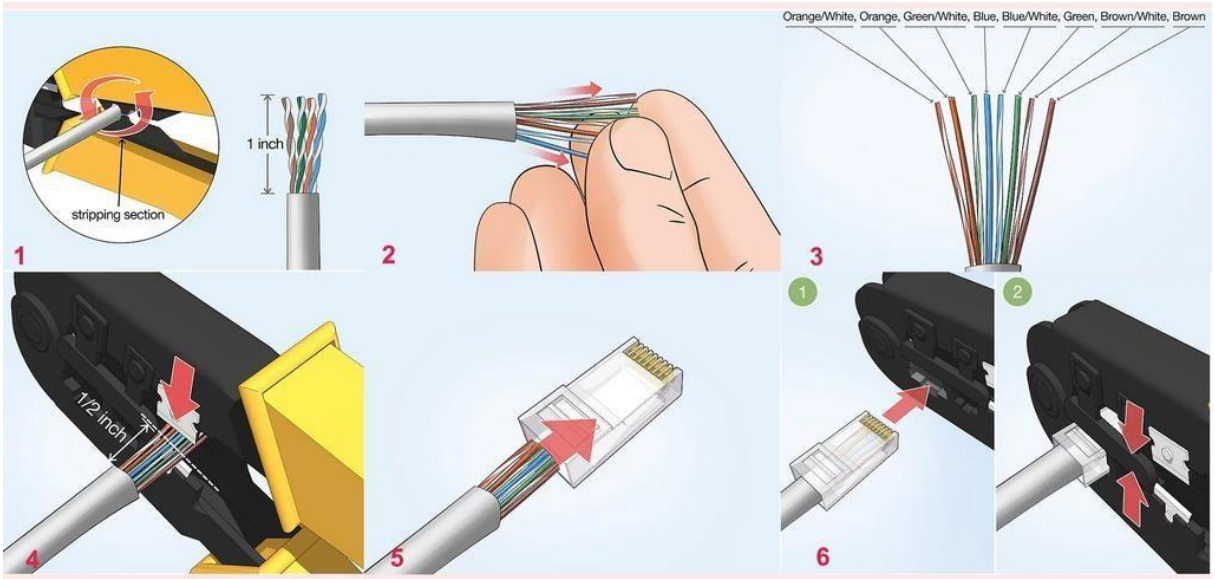
- ST (Straight-Tip) Connector: Features a bayonet (twist-lock) coupling, often used in multimode networks and environments with vibration (like industrial settings)
- FC (Fiber Channel) Connector: Uses a threaded coupling for secure, vibration-resistant connections, common in single-mode applications.
- SC (Subscriber) Connector: Square-shaped, uses a simple push-pull mechanism, common in telecom and datacom for its ease of use and performance.
- LC (Lucent) Connector: Small form factor (SFF), uses a latch mechanism, popular for high-density connections in data centers and telecom due to its small size.

0801CA251105

Q.2) Write process of crimping of LAN cable.

Ans. 2) Crimping of LAN cable is done as per following steps:-

1. Cut the cable to the desired length. Then using the stripper, strip about **1 inch** of the outer jacket off the cable end. And while doing this, be careful not to cut the internal insulation of the internal wires.
2. Unwrap the 4 pairs of twisted wires and separate them and straighten the wires as much as possible.
3. Arrange them in the required order from left to right.
4. Flatten the wires together and trim them to about **1/2 inch** from the edge of the jacket.
5. Insert the wires into the connector, keeping them in order and push until the wires reach the end of the connector.
6. Insert the RJ45 connector into the RJ45 slot of the crimping tool and squeeze the handles of the crimping tool firmly.



0801CA251105

Q.3) What is cross connection and straight connection in networking device.

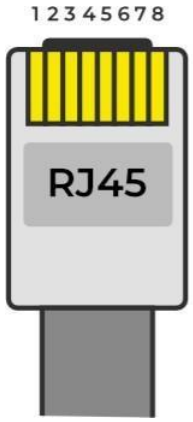
Ans. 3)

- **Straight-Through Connection:**
  - **Wiring:** Pin 1 connects to 1, Pin 2 to 2, etc., on both ends (T568A or T568B used consistently).
  - **Usage:** Connects different types of devices, such as a PC to a switch, or a router to a modem.
  - **Purpose:** The signal sent from the transmitter of one device goes to the receiver of another, and vice-versa.

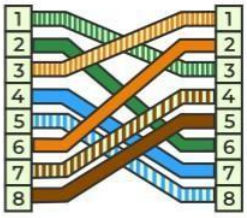
- **Cross Connection (Crossover Cable):**
  - **Wiring:** Pin 1 is crossed with 3, and Pin 2 is crossed with 6 (TX/RX pairs swapped).
  - **Usage:** Connects similar devices directly, such as PC-to-PC, Switch-to-Switch, or Router-to-Router.
  - **Purpose:** Allows direct communication between similar devices, preventing both from trying to transmit on the same line.

Auto MDI-X (Automatic Crossover):

Modern Ethernet ports often support Auto MDI-X, which automatically detects the required cable type and configures the connection internally. This means a straight-through cable can often work for connecting similar devices.



Ethernet Patch Cable



Ethernet Crossover Cable

Assignment-2

S.No.	INDEX	Page No.	Sign
Q	Write the following computer networking commands with their purpose and test run on the application OS :-		
1	ipconfig	1	
2	hostname	1	
3	route print	2	
4	ping	3	
5	nslookup	3	
6	tracert	3	
7	arp -a	4	
8	netstat -a	5	
9	getmac	6	
10	netstat	6	
11	systeminfo	6-7	
12	pathping	8	
13	curl	8	

1. ipconfig : is a **Windows command** used to view and manage your network configuration.

What it shows

- **IPv4 Address** – your local IP
- **Subnet Mask**
- **Default Gateway**
- Network adapters (Wi-Fi, Ethernet, VPNs, etc.)

```
C:\Users\HP>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::fb41:756d:8790:6bce%4
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40c4:1f:1094:74a6:3cee:da93:9912
   Temporary IPv6 Address. . . . . . : 2409:40c4:1f:1094:959e:870d:a13:2f26
   Link-local IPv6 Address . . . . . : fe80::c91:56a8:6046:ff63%14
   IPv4 Address. . . . . . . . . . . : 10.110.172.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::f8a1:69ff:fe9f:54d4%14
                                       10.110.172.251
```

2. hostname : command shows (or sets) the name of your computer on a network.

```
C:\Users\HP>hostname
HP
```

=

3. route print : displays your system's **IP routing table**—basically how your computer decides where network traffic goes.

What it shows

- **Interface List** – network adapters + their indexes
- **IPv4 Route Table / IPv6 Route Table**
    - **Network Destination** – target network
    - **Netmask** – subnet mask
    - **Gateway** – where traffic is sent
    - **Interface** – which adapter is used
    - **Metric** – priority (lower = preferred)

```
C:\Users\HP>route print
===========================================================================
Interface List
  4...0a 00 27 00 00 04 ......VirtualBox Host-Only Ethernet Adapter
 15...1a 47 3d 14 c9 87 ......Microsoft Wi-Fi Direct Virtual Adapter
 12...9a 47 3d 14 c9 87 ......Microsoft Wi-Fi Direct Virtual Adapter #2
 14...18 47 3d 14 c9 87 ......Realtek RTL8822CE 802.11ac PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0   10.110.172.251     10.110.172.1     50
     10.110.172.0    255.255.255.0         On-link      10.110.172.1    306
     10.110.172.1  255.255.255.255         On-link      10.110.172.1    306
   10.110.172.255  255.255.255.255         On-link      10.110.172.1    306
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      192.168.56.0    255.255.255.0         On-link      192.168.56.1    281
      192.168.56.1  255.255.255.255         On-link      192.168.56.1    281
    192.168.56.255  255.255.255.255         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link      10.110.172.1    306
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link      192.168.56.1    281
  255.255.255.255  255.255.255.255         On-link      10.110.172.1    306
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
 14     66 ::/0                     fe80::f8a1:69ff:fe9f:54d4
  1    331 ::1/128                  On-link
 14     66 2409:40c4:1f:1094::/64   On-link
 14    306 2409:40c4:1f:1094:74a6:3cee:da93:9912/128
                                    On-link
 14    306 2409:40c4:1f:1094:959e:870d:a13:2f26/128
                                    On-link
  4    281 fe80::/64                On-link
 14    306 fe80::/64                On-link
 14    306 fe80::c91:56a8:6046:ff63/128
                                    On-link
  4    281 fe80::fb41:756d:8790:6bce/128
                                    On-link
  1    331 ff00::/8                 On-link
  4    281 ff00::/8                 On-link
 14    306 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

=

4. ping : command tests **network connectivity** between your computer and another host.

```
C:\Users\HP>ping google.com

Pinging google.com [2404:6800:4009:80c::200e] with 32 bytes of data:
Reply from 2404:6800:4009:80c::200e: time=39ms
Reply from 2404:6800:4009:80c::200e: time=67ms
Reply from 2404:6800:4009:80c::200e: time=47ms
Reply from 2404:6800:4009:80c::200e: time=43ms

Ping statistics for 2404:6800:4009:80c::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 39ms, Maximum = 67ms, Average = 49ms
```

5. nslookup : is a command used to **query DNS** (Domain Name System) and resolve **domain names ↔ IP addresses**.

```
C:\Users\HP>nslookup google.com
Server:  UnKnown
Address:  10.110.172.251

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4009:81e::200e
          142.250.67.238
```

6. tracert : is a command used to **trace the path** packets take from your computer to a destination host.

What it shows

Each line = a **hop** (router) between you and the destination:

- Hop number
- Round-trip time (ms)
- Router IP / hostname

```
C:\Users\HP>tracert google.com

Tracing route to google.com [2404:6800:4009:81e::200e]
over a maximum of 30 hops:

  1     5 ms     5 ms     4 ms  2409:40c4:1f:1094::a5
  2     *        *        *     Request timed out.
  3    26 ms    38 ms    41 ms  2405:200:5207:21:3925::1
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    66 ms    30 ms    39 ms  2405:200:801:1200::10fe
  7     *        *        *     Request timed out.
  8    85 ms    79 ms    71 ms  2405:200:100::1:0:5c8
  9     *        *        *     Request timed out.
 10   108 ms    86 ms    50 ms  2404:6800:80b1::1
 11    69 ms    41 ms    79 ms  2001:4860:0:1::11c2
 12    50 ms    77 ms    50 ms  2001:4860:0:1::40a1
 13    75 ms    78 ms    77 ms  lcboma-ba-in-x0e.1e100.net [2404:6800:4009:81e::200e]

Trace complete.
```

=

7. arp -a : displays the **ARP cache** — a table that maps **IP addresses to MAC addresses** on your local network.

What it shows

For each entry:

- **Internet Address** → IP address
- **Physical Address** → MAC address
- **Type** → dynamic or static

```
C:\Users\HP>arp -a

Interface: 192.168.56.1 --- 0x4
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.5             01-00-5e-00-00-05     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.1.60            01-00-5e-00-01-3c     static
  224.0.2.3             01-00-5e-00-02-03     static
  224.0.2.4             01-00-5e-00-02-04     static
  224.2.2.2             01-00-5e-02-02-02     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  239.255.255.253       01-00-5e-7f-ff-fd     static

Interface: 10.110.172.1 --- 0xe
  Internet Address      Physical Address      Type
  10.110.172.251        fa-a1-69-9f-54-d4     dynamic
  10.110.172.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

=

8. netstat -a : shows **all network connections and listening ports** on your system

```
C:\Users\HP>netstat -a

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            HP:0                   LISTENING
  TCP    0.0.0.0:445            HP:0                   LISTENING
  TCP    0.0.0.0:5040           HP:0                   LISTENING
  TCP    0.0.0.0:5432           HP:0                   LISTENING
  TCP    0.0.0.0:7680           HP:0                   LISTENING
  TCP    0.0.0.0:49664          HP:0                   LISTENING
  TCP    0.0.0.0:49665          HP:0                   LISTENING
  TCP    0.0.0.0:49666          HP:0                   LISTENING
  TCP    0.0.0.0:49667          HP:0                   LISTENING
  TCP    0.0.0.0:49668          HP:0                   LISTENING
  TCP    0.0.0.0:49670          HP:0                   LISTENING
  TCP    10.110.172.1:139       HP:0                   LISTENING
  TCP    10.110.172.1:54997     4.213.25.242:https     ESTABLISHED
  TCP    10.110.172.1:64989     104.208.16.92:https    TIME_WAIT
  TCP    127.0.0.1:15292        HP:0                   LISTENING
  TCP    127.0.0.1:15393        HP:0                   LISTENING
  TCP    127.0.0.1:16494        HP:0                   LISTENING
  TCP    127.0.0.1:45623        HP:0                   LISTENING
  TCP    127.0.0.1:53738        HP:0                   LISTENING
  TCP    127.0.0.1:61331        HP:0                   LISTENING
  TCP    127.0.0.1:61332        HP:0                   LISTENING
  TCP    192.168.56.1:139       HP:0                   LISTENING
  TCP    [::]:135               HP:0                   LISTENING
  TCP    [::]:445               HP:0                   LISTENING
  TCP    [::]:5432              HP:0                   LISTENING
  TCP    [::]:7680              HP:0                   LISTENING
  TCP    [::]:49664             HP:0                   LISTENING
  TCP    [::]:49665             HP:0                   LISTENING
  TCP    [::]:49666             HP:0                   LISTENING
  TCP    [::]:49667             HP:0                   LISTENING
  TCP    [::]:49668             HP:0                   LISTENING
  TCP    [::]:49670             HP:0                   LISTENING
  TCP    [::1]:42050            HP:0                   LISTENING
  TCP    [::1]:49669            HP:0                   LISTENING
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:54998  [2603:1040:a06:6::2]:https  ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:55499  sm-in-f188:5228        ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:61343  ec2-54-178-125-94:https  ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:61348  ec2-52-73-45-44:https  ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:64960  ec2-176-34-53-12:https  ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:64968  ec2-3-231-72-45:https  ESTABLISHED
  UDP    0.0.0.0:5050           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5355           *:*
  UDP    0.0.0.0:52198          *:*
  UDP    0.0.0.0:57739          *:*
  UDP    0.0.0.0:61821          0.0.32.4:443
  UDP    10.110.172.1:137       *:*
  UDP    10.110.172.1:138       *:*
  UDP    10.110.172.1:1900      *:*
  UDP    10.110.172.1:57282     *:*
  UDP    127.0.0.1:1900         *:*
  UDP    127.0.0.1:56354        127.0.0.1:56354
  UDP    127.0.0.1:57283        *:*
  UDP    127.0.0.1:60435        127.0.0.1:60436
  UDP    127.0.0.1:60436        127.0.0.1:60435
  UDP    192.168.56.1:137       *:*
  UDP    192.168.56.1:138       *:*
  UDP    192.168.56.1:1900      *:*
  UDP    192.168.56.1:57281     *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5355              *:*
  UDP    [::]:52198             *:*
  UDP    [::]:57739             *:*
  UDP    [::]:61821             [2404:6800:4009:807::2004]:443
  UDP    [::1]:1900             *:*
  UDP    [::1]:57280            *:*
  UDP    [fe80::c91:56a8:6046:ff63%14]:1900  *:*
  UDP    [fe80::c91:56a8:6046:ff63%14]:57279  *:*
  UDP    [fe80::fb41:756d:8790:6bce%4]:1900  *:*
  UDP    [fe80::fb41:756d:8790:6bce%4]:57278  *:*
```

=

9. getmac : displays the **MAC (physical) address** of your network adapters.

```
C:\Users\HP>getmac

Physical Address    Transport Name
=================== ==========================================================
18-47-3D-14-C9-87   \Device\Tcpip_{B52890CA-44BE-4AFE-B96A-6515E3110F15}
0A-00-27-00-00-04   \Device\Tcpip_{0C5E7FD5-16A6-4AED-8E0E-886D570CFDE2}
```

10. netstat : (short for **network statistics**) is a command that shows **active network connections, listening ports, and network protocol statistics**.

```
C:\Users\HP>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.110.172.1:54997     4.213.25.242:https     ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:54998  [2603:1040:a06:6::2]:https  ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:55499  sm-in-f188:5228        ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:61343  ec2-54-178-125-94:https  ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:61348  ec2-52-73-45-44:https  ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:64960  ec2-176-34-53-12:https  ESTABLISHED
  TCP    [2409:40c4:1f:1094:959e:870d:a13:2f26]:65010  ec2-34-203-175-187:https  ESTABLISHED
```

11. systeminfo : command displays **detailed information about your Windows computer and operating system**.

What it shows

- **OS Name & Version**
- **System Manufacturer & Model**
- **Processor(s)**
- **BIOS Version**
- **Installed RAM**
- **Network Adapter(s)**
- **Hotfixes / Updates Installed**
- **System Boot Time**

=

```
C:\Users\HP>systeminfo

Host Name:                     HP
OS Name:                       Microsoft Windows 11 Home Single Language
OS Version:                    10.0.26200 N/A Build 26200
OS Manufacturer:               Microsoft Corporation
OS Configuration:              Standalone Workstation
OS Build Type:                 Multiprocessor Free
Registered Owner:              HP
Registered Organization:
Product ID:                    00327-35912-46760-AAOEM
Original Install Date:         9/14/2025, 1:51:12 AM
System Boot Time:              1/21/2026, 11:18:22 AM
System Manufacturer:           HP
System Model:                  HP Laptop 14s-dr2xxx
System Type:                   x64-based PC
Processor(s):                  1 Processor(s) Installed.
                               [01]: Intel64 Family 6 Model 140 Stepping 1 GenuineIntel ~2401 Mhz
BIOS Version:                  AMI F.33, 10/4/2023
Windows Directory:             C:\WINDOWS
System Directory:              C:\WINDOWS\system32
Boot Device:                   \Device\HarddiskVolume1
System Locale:                 en-us;English (United States)
Input Locale:                  en-us;English (United States)
Time Zone:                     (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:         7,835 MB
Available Physical Memory:     875 MB
Virtual Memory: Max Size:      13,894 MB
Virtual Memory: Available:     711 MB
Virtual Memory: In Use:        13,183 MB
Page File Location(s):         C:\pagefile.sys
Domain:                        WORKGROUP
Logon Server:                  \\HP
Hotfix(s):                     4 Hotfix(s) Installed.
                               [01]: KB5066128
                               [02]: KB5054156
                               [03]: KB5074109
                               [04]: KB5071142
Network Card(s):               2 NIC(s) Installed.
                               [01]: Realtek RTL8822CE 802.11ac PCIe Adapter
                                     Connection Name: Wi-Fi
                                     DHCP Enabled:    Yes
                                     DHCP Server:     10.110.172.251
                                     IP address(es)
                                     [01]: 10.110.172.1
                                     [02]: fe80::c91:56a8:6046:ff63
                                     [03]: 2409:40c4:1f:1094:959e:870d:a13:2f26
                                     [04]: 2409:40c4:1f:1094:74a6:3cee:da93:9912
                               [02]: VirtualBox Host-Only Ethernet Adapter
                                     Connection Name: Ethernet
                                     DHCP Enabled:    No
                                     IP address(es)
                                     [01]: 192.168.56.1
                                     [02]: fe80::fb41:756d:8790:6bce
Virtualization-based security: Status: Running
                               Required Security Properties:
                               Available Security Properties:
                                     Base Virtualization Support
                                     DMA Protection
                                     UEFI Code Readonly
                                     SMM Security Mitigations 1.0
                                     Mode Based Execution Control
                                     APIC Virtualization
                               Services Configured:
                                     Hypervisor enforced Code Integrity
                               Services Running:
                                     Hypervisor enforced Code Integrity
                               App Control for Business policy: Enforced
                               App Control for Business user mode policy: Off
                               Security Features Enabled:
Hyper-V Requirements:          A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```

=

12. pathping : is a Windows command that combines the functionality of **ping** and **tracert**. It shows the **route to a host** and **packet loss at each hop** along the way.

What it does

1. Traces the **path (hops) to the destination**, like tracert.
2. Sends multiple **pings to each hop** over time.
3. Shows **packet loss % and latency** for each hop.

```
C:\Users\HP>pathping google.com

Tracing route to google.com [2404:6800:4002:828::200e]
over a maximum of 30 hops:
  0  HP [2409:40c4:1f:1094:959e:870d:a13:2f26]
  1  2409:40c4:1f:1094::a5
  2     *        *        *
Computing statistics for 25 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           HP [2409:40c4:1f:1094:959e:870d:a13:2f26]
                                 0/ 100 =  0%   |
  1    9ms     0/ 100 =  0%     0/ 100 =  0%  2409:40c4:1f:1094::a5

Trace complete.
```

13. curl : is a command-line tool used to **transfer data to or from a server** using various protocols like **HTTP, HTTPS, FTP, and more**.

```
C:\Users\HP>curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```

=

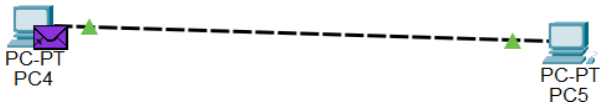
Assignment-3

S.No.	INDEX	Page No.	Sign
1	Simulation of peer to peer connection using packet tracer.	1	
2	Simulation of star topology with hub and switch using packet tracer.	2-3	

Q.1) Simulation of peer to peer connection using packet tracer.

Ans. 1) Simulation of peer to peer connection is done as per the following steps:-

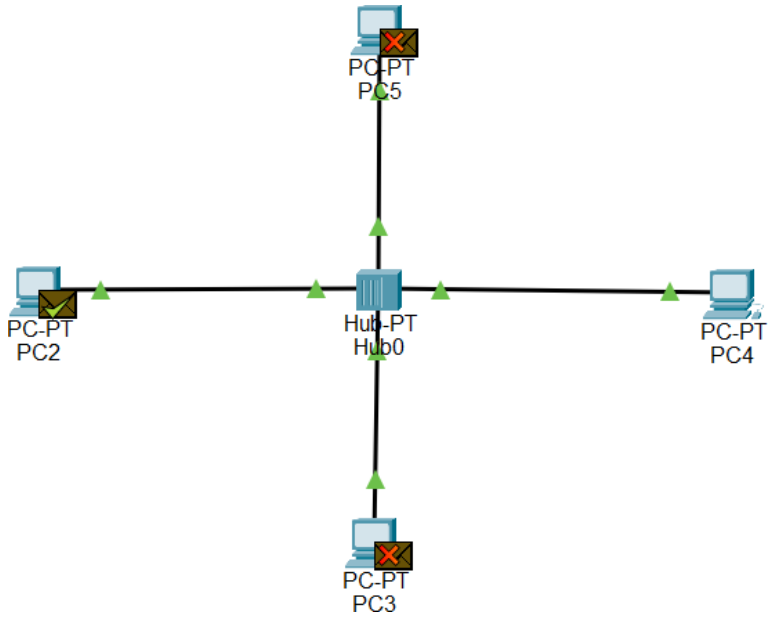
1. Select two PC's by PC button from End Devices button.
2. Connect these two PC's by Copper Cross-Over button from Connections button and by selecting FastEthernet ports in both.
3. Set IPv4 addresses on both PC's by clicking on PC -> Desktop -> Ipconfiguration. (10.10.10.1 on 1st PC and 10.10.10.2 on 2nd PC, subnet mask – 255.0.0.0 (will autofill as per class)
4. Send message from 1st PC to 2nd PC by clicking on Add Simple PDU (P), 1st click on sender and 2nd click on receiver.
5. Click on the simulation button and play the message transfer process.

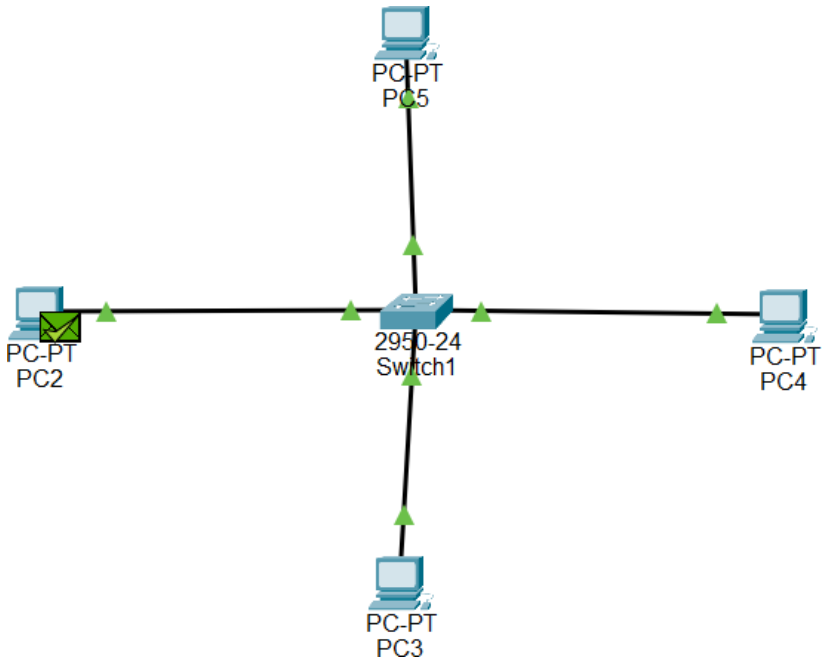


Q.2) Simulation of star topology with hub and switch using packet tracer.

Ans.2) Steps to Simulate Star Topology :-

1. Add a Switch (e.g., 2950-24) or Hub from "Network Devices" and multiple PCs from "End Devices" onto the workspace.
2. Use "Copper Straight-Through" cables to connect the FastEthernet port of each PC to any available port on the central switch/hub.
3. Click each PC, go to the "Desktop" tab, select "IP Configuration," and assign an IPv4 address (e.g., 10.10.10.1, 10.10.10.2, etc.). subnet mask – 255.0.0.0 (will autofill as per class)
4. Verification (if required): Open the command prompt on one PC and type ping [IP address of destination PC] to check connectivity.
5. Send message from 1st PC to 2nd PC by clicking on Add Simple PDU (P), 1st click on sender and 2nd click on receiver.
6. **Observation:** In simulation mode, a switch will show packets going only to the destination, while a hub will show packets being broadcast to all ports.




PC-PT
PC5
PC-PT
PC2
2950-24
Switch1
PC-PT
PC4
PC-PT
PC3